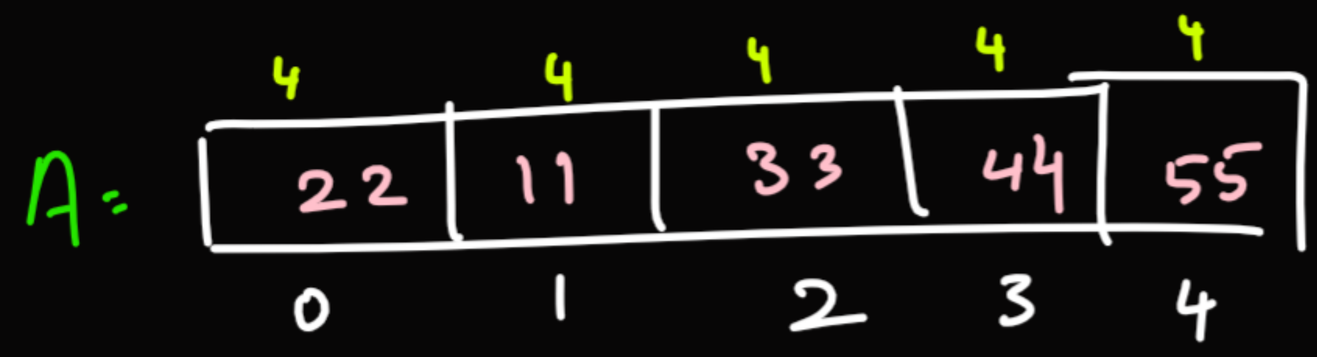
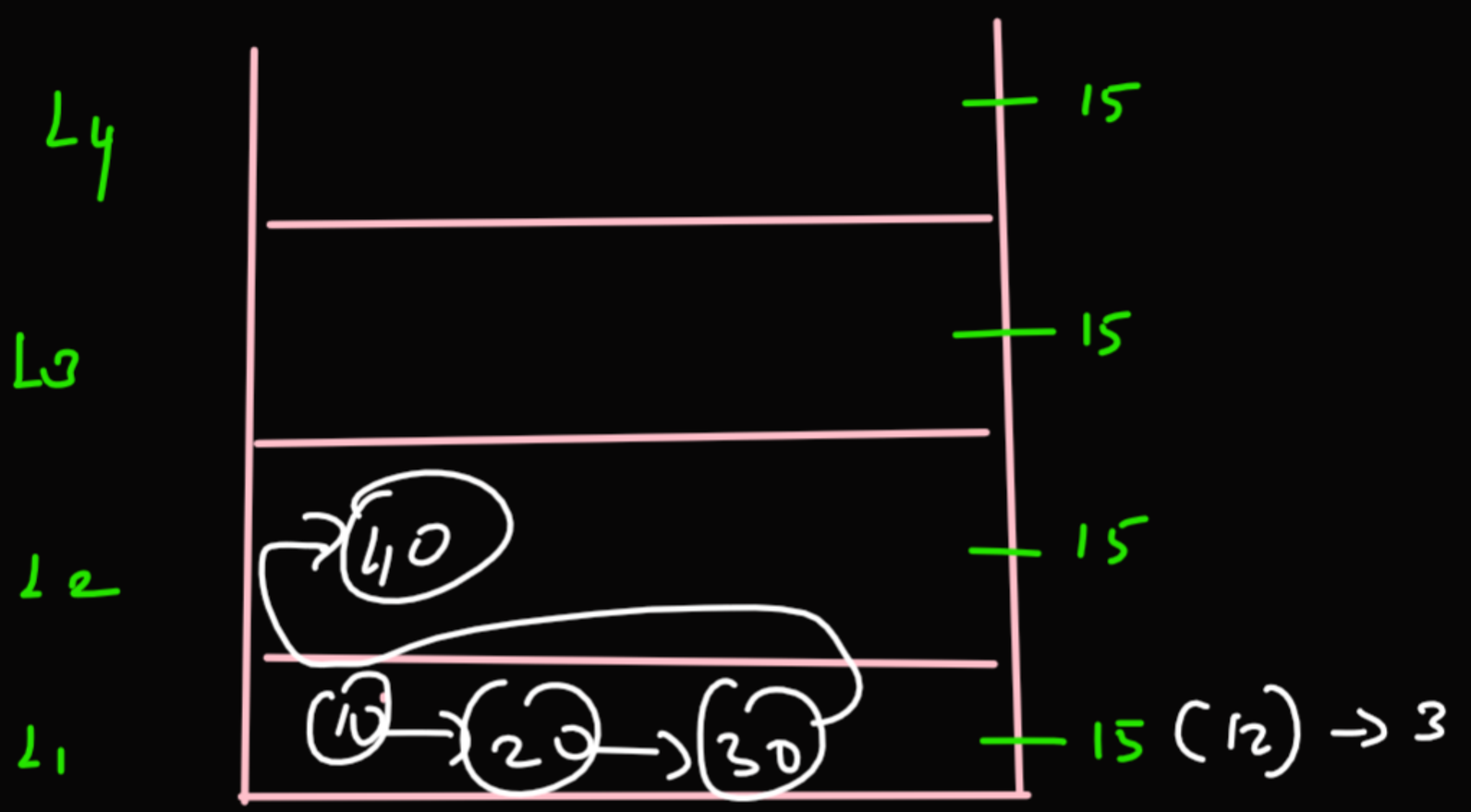
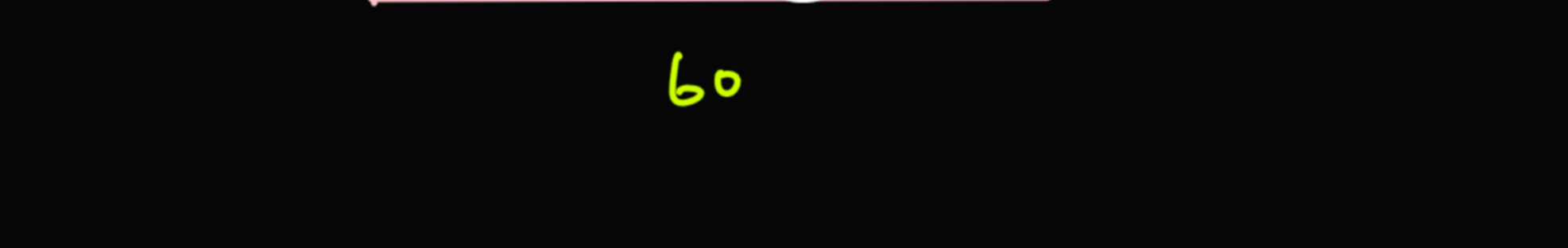


= 20 bytes

A: 20



60



" " String → group of character

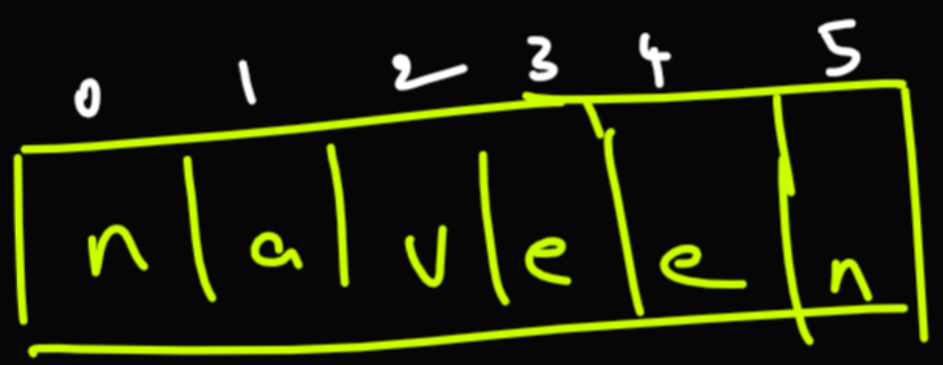
Immutable (can't alter & update)

String Str = "naveen";

Scanner Scn = new Scanner(System.in);

Str = Scn.next(); welcome To At

Scn.nextLine();



Str = " n a v e e n "
0 1 2 3 4 5

Str.charAt(3) - e

A[3] = e

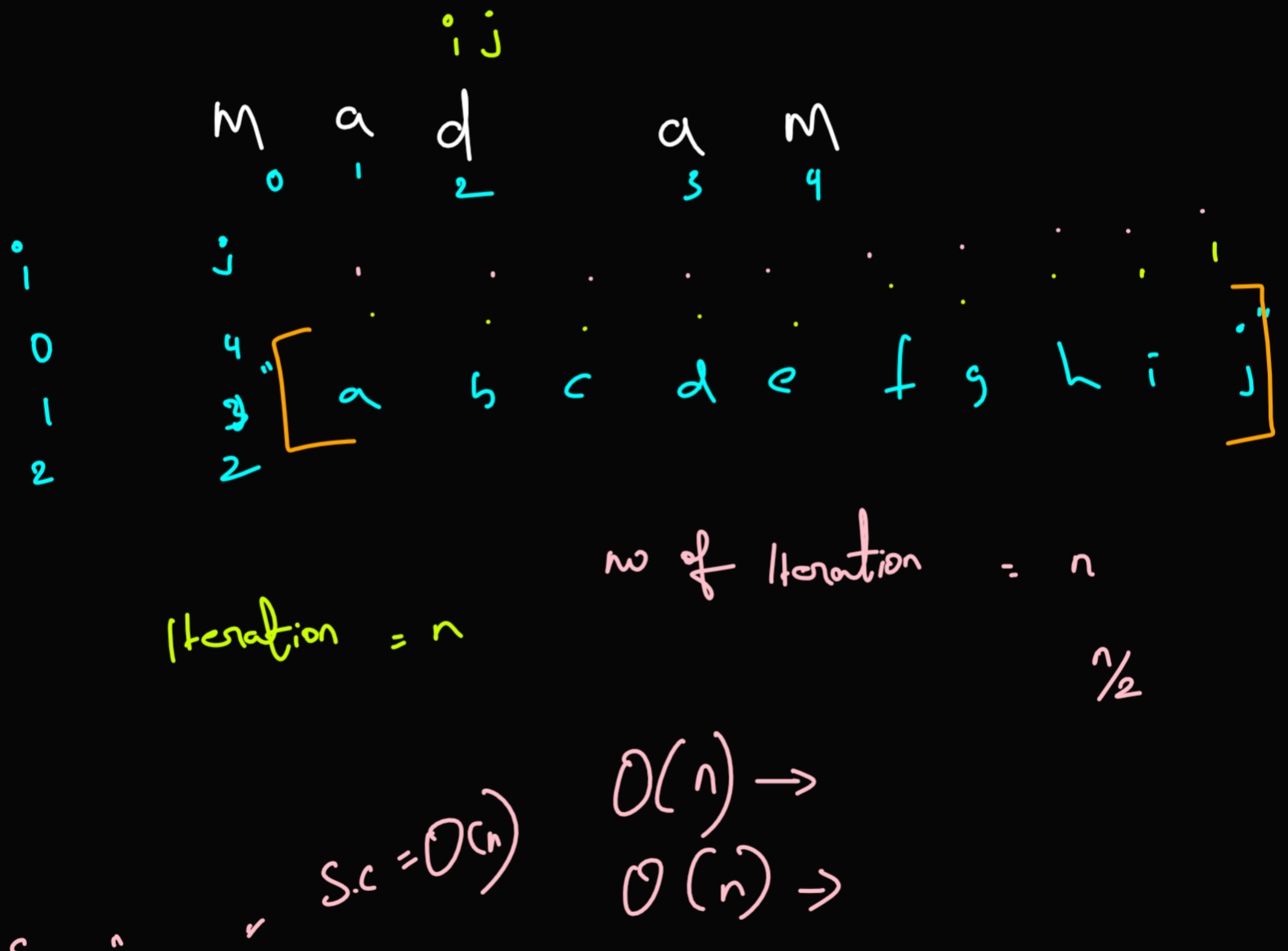
Str.Substring(st, end)
inc, 1
exc, 5

String Palindrome

S1 = "racecar"

S2 = ""

S2 = racecar



Iteration = n

no of Iteration = n

n/2

S.C = O(n)

O(n) →

O(n) →

S1 = "

S2 =

S.C = O(1)

Leetcode 2108

```java
class Solution {
    public String firstPalindrome(String[] words) {
        for(String word:words){
            if(isPalindrome(word)){
                return word;
            }
        }
        return "";
    }
    boolean isPalindrome(String A){
        int i=0;
        int j=A.length()-1;
        while(i<j){
            if(A.charAt(i)!=A.charAt(j))
                return false;

            i++;
            j--;
        }
        return true;
    }
```

Search in a row-col Sorted matrix

	0	1	2
0	3	30	38
1	20	52	54
2	35	60	69

n x m

while (r ≥ 0 && r < n && c ≥ 0 && c < m)

2

62

r	c
0	2
1	2
2	2
2	1

if (pos < x)
r++;
if (pos > x)
c--;
if (pos == x)

```java
// your code here
int n=A.length;
int m=A[0].length;
int r=0;
int c=m-1;
while(r>=0 && r<n && c>=0 && c<m){
    if(A[r][c]<x){
        r++;
    }else if(A[r][c]>x){
        c--;
    }else{
        return true;
    }
}
return false;
```

Char & Strings Brush up

char ch = 'a',

isUpperCase

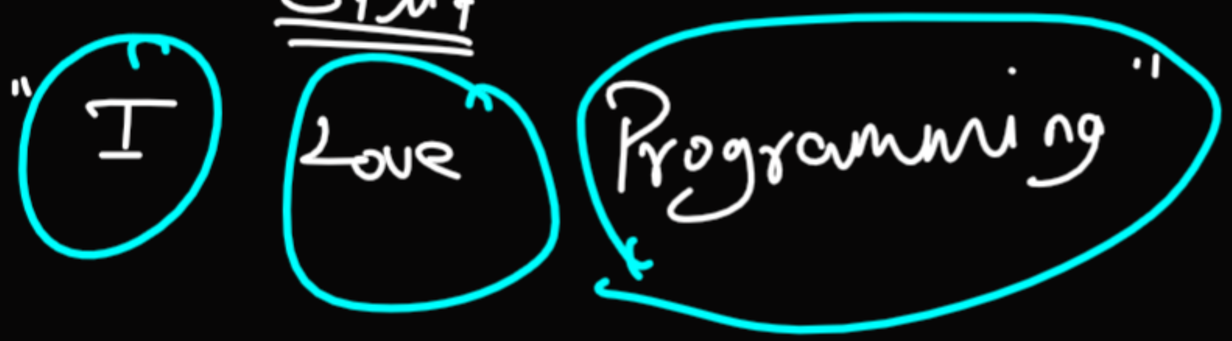
String str = " "

```java
// your code here
int uc=0;
int lc=0;
int d=0;
int sc=0;
for(int i=0;i<s.length();i++){
    char ch=s.charAt(i);
    if(Character.isUpperCase(ch)){
        uc++;
    }else if(Character.isLowerCase(ch)){
        lc++;
    }else if(Character.isDigit(ch)){
        d++;
    }else{
        sc++;
    }
}
int ans[]=new int[4];
ans[0]=uc;
ans[1]=lc;
ans[2]=d;
ans[3]=sc;
return ans;
}
```

String & methods

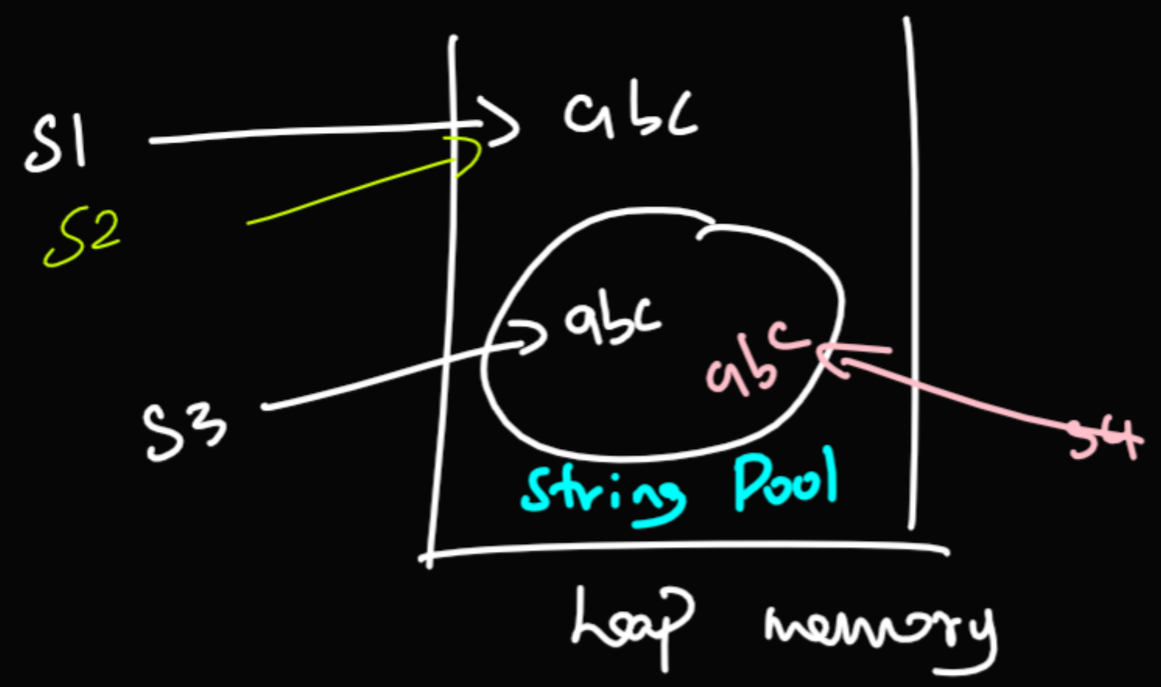
1) Group of characters, Immutable.

1) charAt → char
2) length → Integer
3) substring (st, end) → String
4) substring (st) → String
5) replace (s, T) → new String with updation
6) replaceAll ("s", "T") → Unns
7) toCharArray() → char Array
8) split (" ") → condition
9) s1.equals(s2) → true / false } bool



String Arr[] = { "I", "Love", "Programming" }

```
String str="I Love Programming";
System.out.println(str.length());
System.out.println(str.substring(3));
System.out.println(str.replaceAll("[ao]",""));
char[] Arr=str.toCharArray();
String[] splitArr=str.split(" ");
System.out.println(Arr[0]);
```

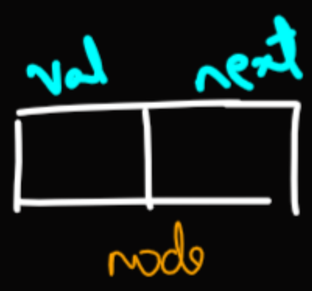


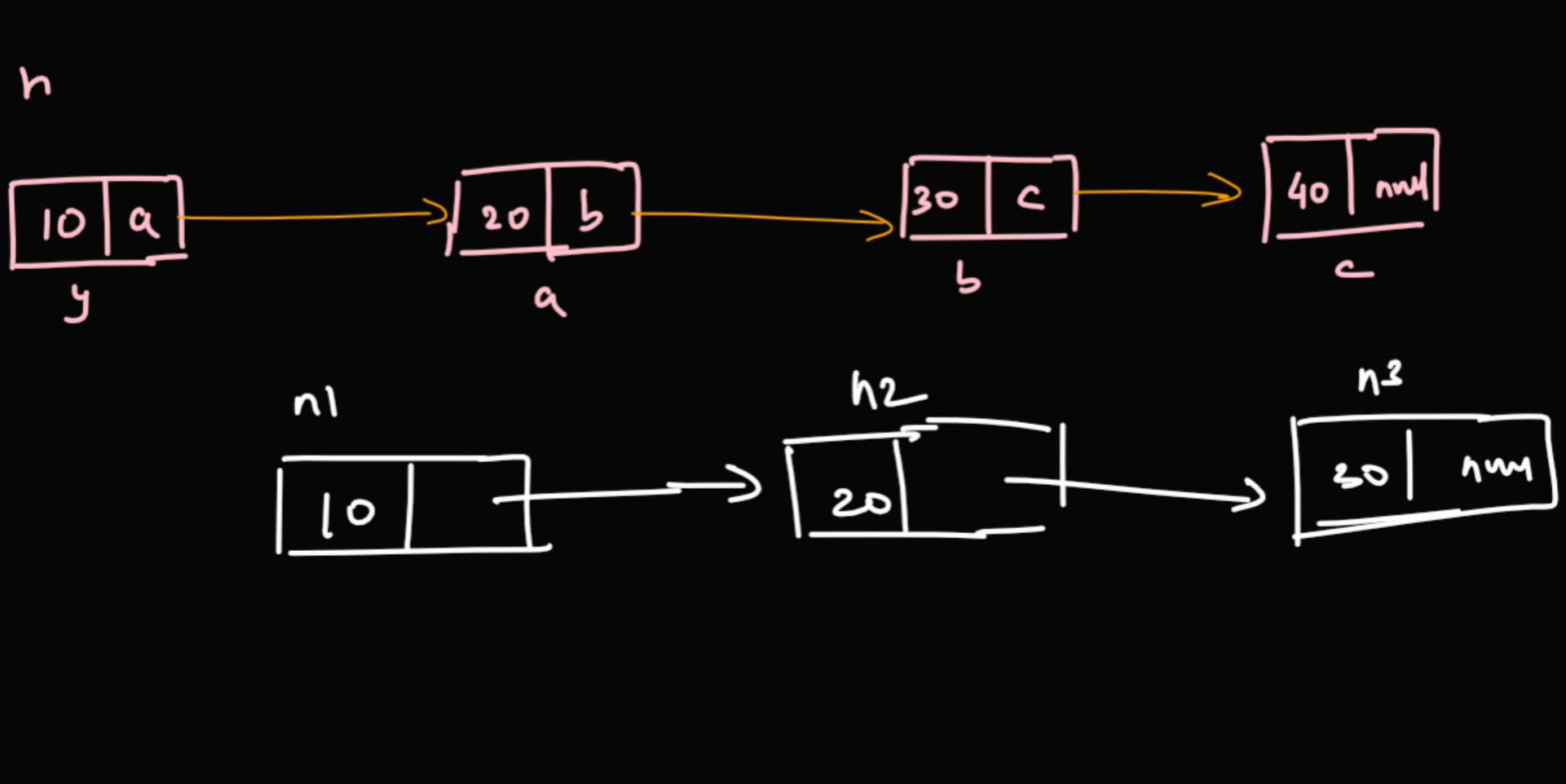
== (to compare Address) s1.equals(s2) - To compare values

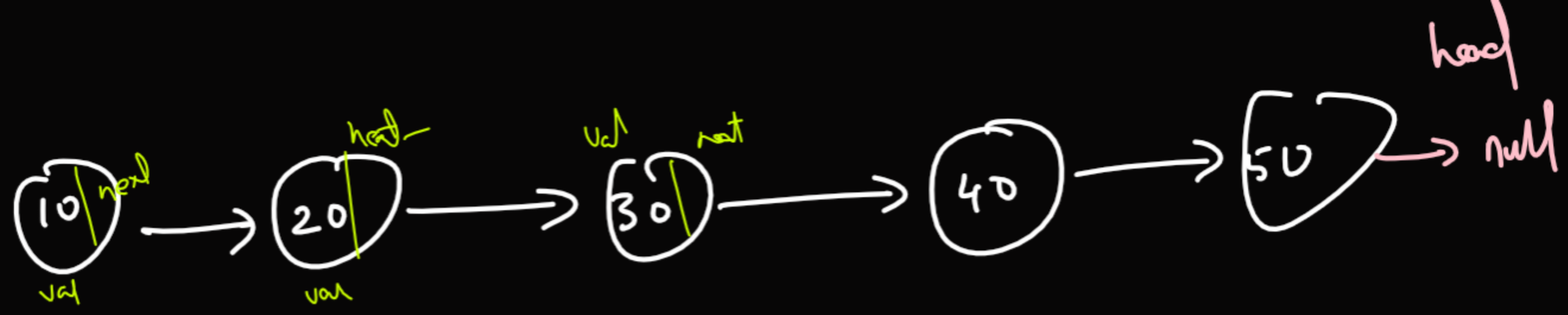
class Intro

```
class student{
    int age;
    String name;
    student(int age,String name){
        this.age=age;
        this.name=name;
    }
}
```

Linked List








```
while(head!=null){
    Print(head.val) —— 10,20
    head=head.next
}
```

3

Display a Linked list

```java
class Solution {
    // Function to display the elements of a li
    void printList(Node head) {
        // add code here.
        while(head!=null){
            System.out.print(head.data+" ");
            head=head.next;
        }
    }
}
```